

2 (a) Researchers have found evidence of natural selection in humans.

- Originally, in human populations it was only babies and children that needed to digest the milk sugar, lactose. The gene coding for the enzyme lactase (*LCT* gene) was switched off before adulthood.
- Today, in many populations, some adult individuals have lactose intolerance, which means they cannot digest lactose. Lactose intolerance leads to side-effects such as abdominal pain after eating food containing lactose.
- A mutation has been identified that keeps the *LCT* gene switched on. An adult who has this mutation is able to digest lactose. This is called lactose persistence.
- Lactose persistence increased in populations in Europe several thousand years ago.
- The increase in lactose persistence in Europe coincided with an increase in farming of cows for milk.

(i) Natural selection has caused this increase in lactose persistence.

State the type of selection that has caused this increase.

..... [1]

(ii) Explain why there was selection for lactose persistence in humans several thousand years ago.

.....
.....
.....
.....
.....
.....
..... [3]

(b) Lactose intolerance and lactose persistence were investigated in a test population in Europe. The mutation which causes lactose persistence is in a regulatory gene (**T/t**).

- People with lactose intolerance have the genotype **tt**.
- People with lactose persistence have the genotypes **TT** and **Tt**.
- 166 people were tested for their genotype.
- 58 people were found to have lactose intolerance.

- (i) The Hardy–Weinberg principle can be used to calculate allele, genotype and phenotype frequencies in populations.

The Hardy–Weinberg equations are shown in Fig. 2.1:

$$p + q = 1$$
$$p^2 + 2pq + q^2 = 1$$

Fig. 2.1

Calculate the frequency of allele **T**.

Show your working.

frequency of allele **T** = [3]

- (ii) When the calculated phenotype frequencies were compared to those in the general population in Europe, it was found that the percentage of people with lactose intolerance in this test population was much higher than in the general population.

Suggest **two** reasons why the percentage of people with lactose intolerance was much higher in the test population than in the general population.

.....
.....
.....
..... [2]

- (c) In eukaryotes, gene expression is controlled by transcription factors, coded for by regulatory genes.

- (i) Outline ways in which transcription factors carry out their role.

.....
.....
.....
..... [2]

- (ii) It is estimated that 2% of human DNA consists of genes coding for proteins (structural genes). Of the remaining 98%, some of the DNA consists of regulatory genes and control sequences that together control gene expression.

State **one** type of control sequence found in human DNA.

..... [1]

- (iii) A study of human evolution identified the location of mutations that result in a change in human phenotype. The study found most examples of mutations had occurred in regulatory genes, not structural genes.

Suggest **and** explain why most changes in human phenotype are due to mutations in regulatory genes.

.....
.....
.....
.....
..... [2]

[Total: 14]